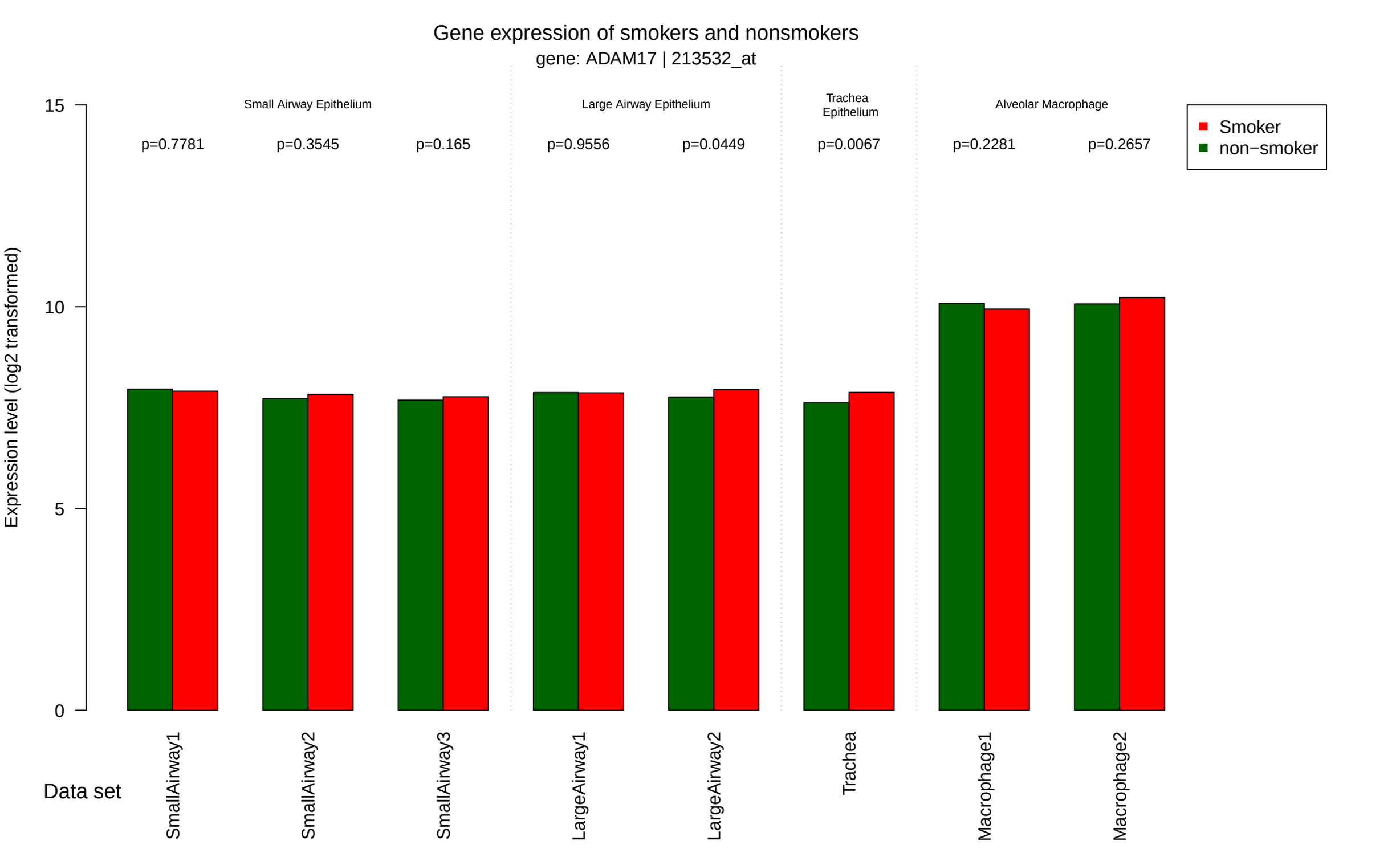
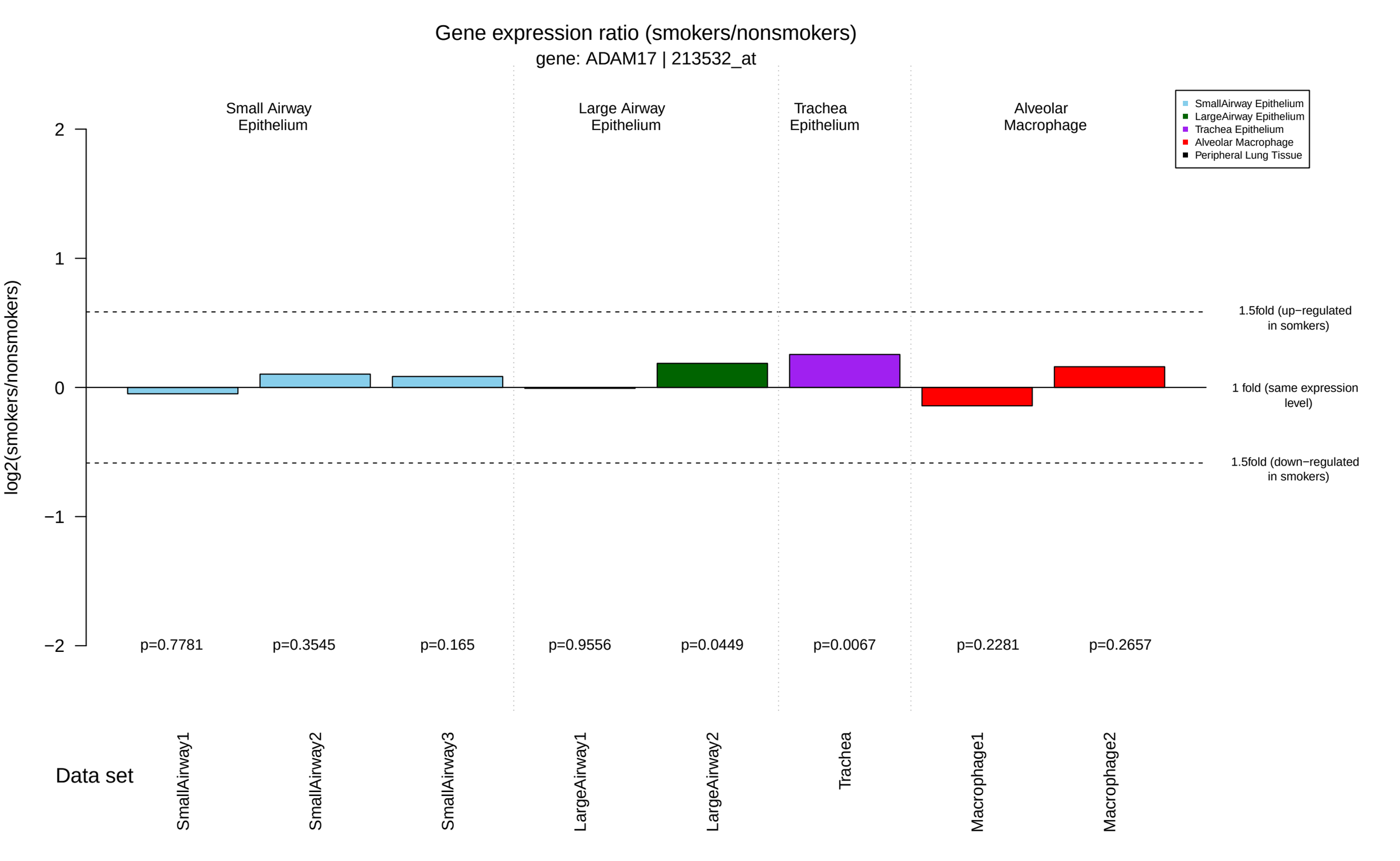
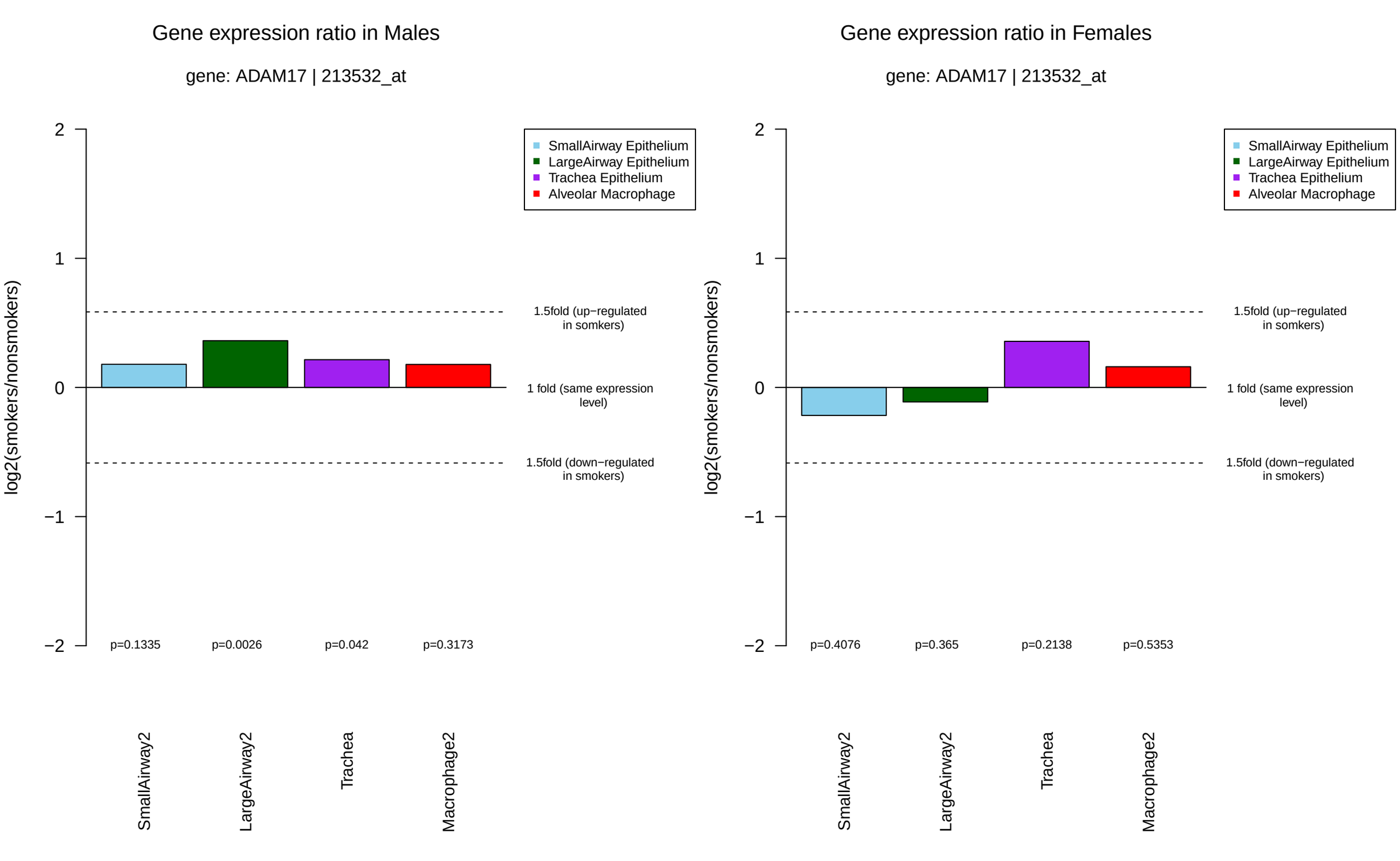
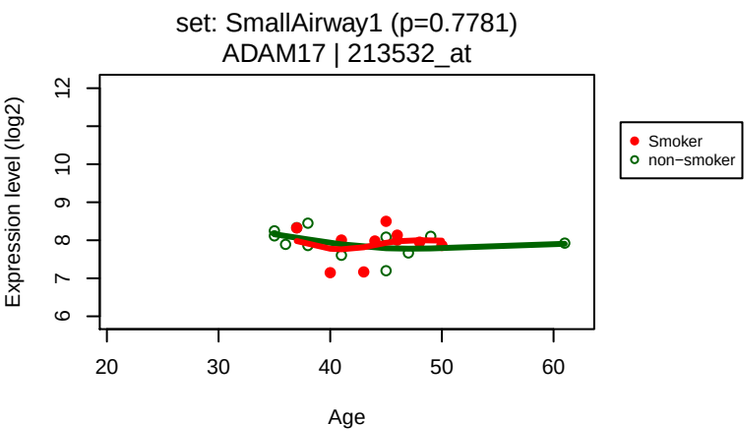
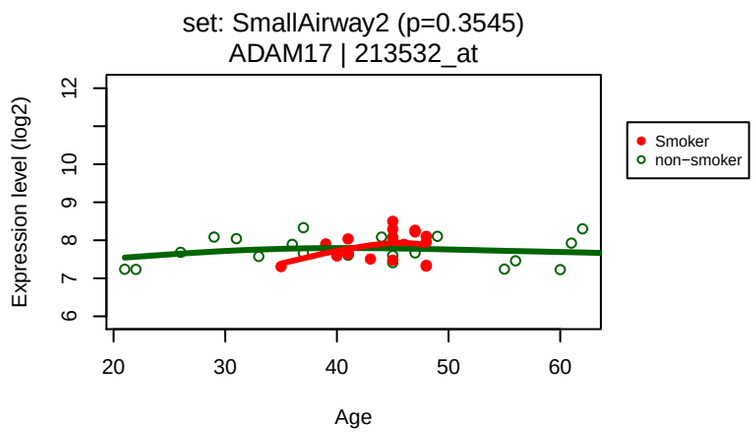
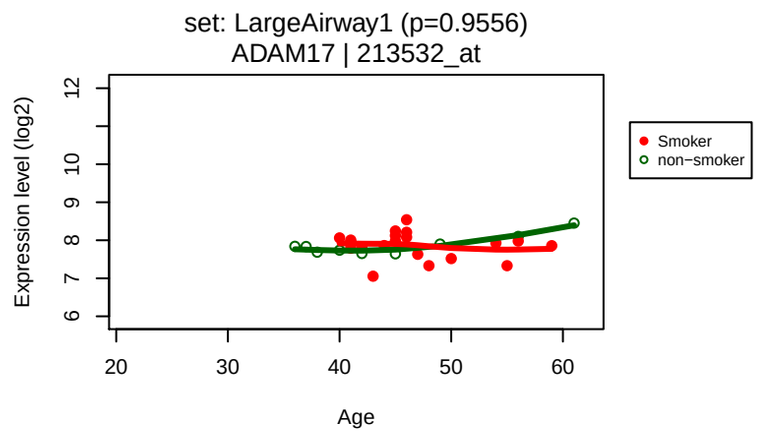

Gene expression of smokers and nonsmokers
gene: ADAM17 | 213532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7781
p=0.3545
p=0.165
p=0.9556
p=0.0449
p=0.0067
p=0.2281
p=0.2657
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

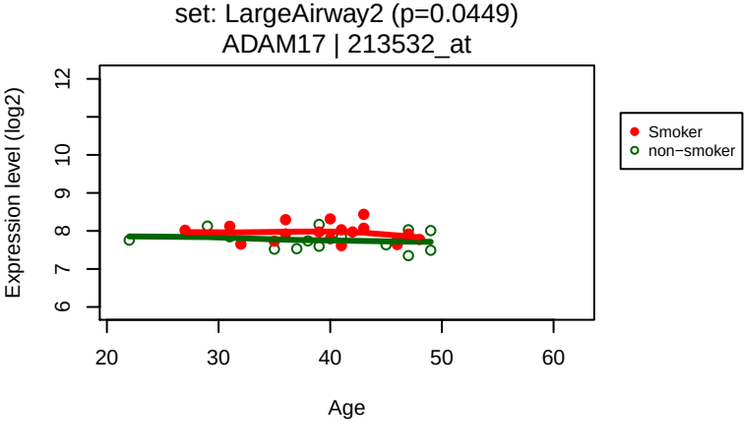

Gene expression ratio (smokers/nonsmokers)
gene: ADAM17 | 213532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7781
p=0.3545
p=0.165
p=0.9556
p=0.0449
p=0.0067
p=0.2281
p=0.2657
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

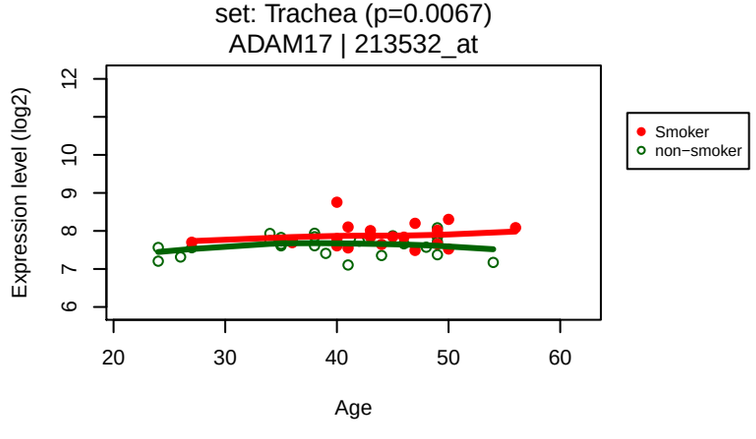

Gene expression ratio in Males
gene: ADAM17 | 213532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1335
p=0.0026
p=0.042
p=0.3173
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAM17 | 213532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4076
p=0.365
p=0.2138
p=0.5353
SmallAirway2
LargeAirway2
Trachea
Macrophage2

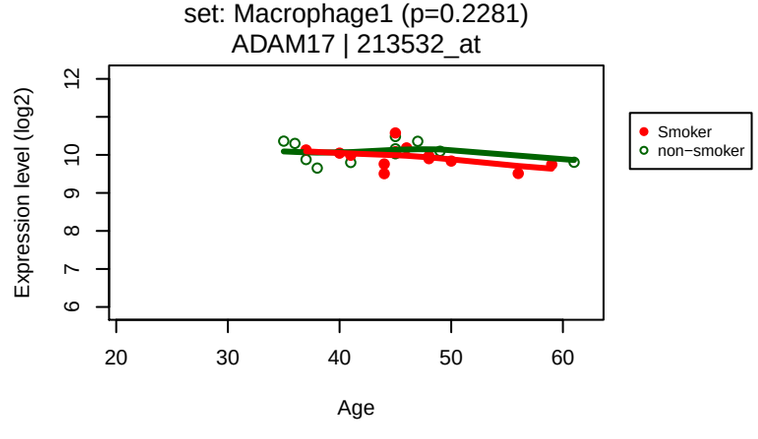

set: SmallAirway1 (p=0.7781)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker

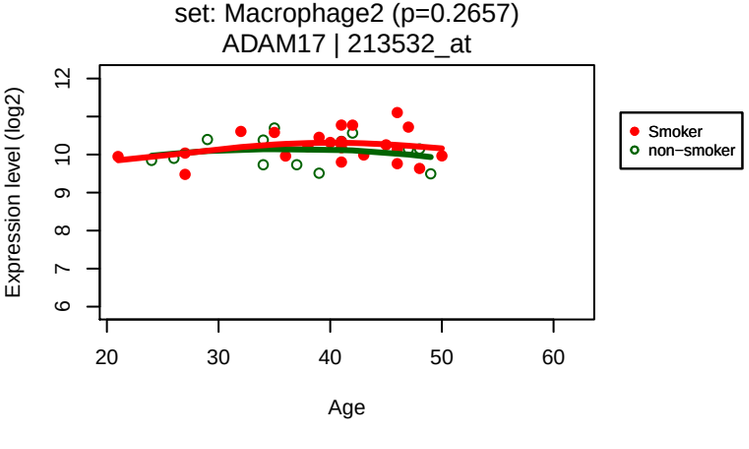

set: SmallAirway2 (p=0.3545)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker

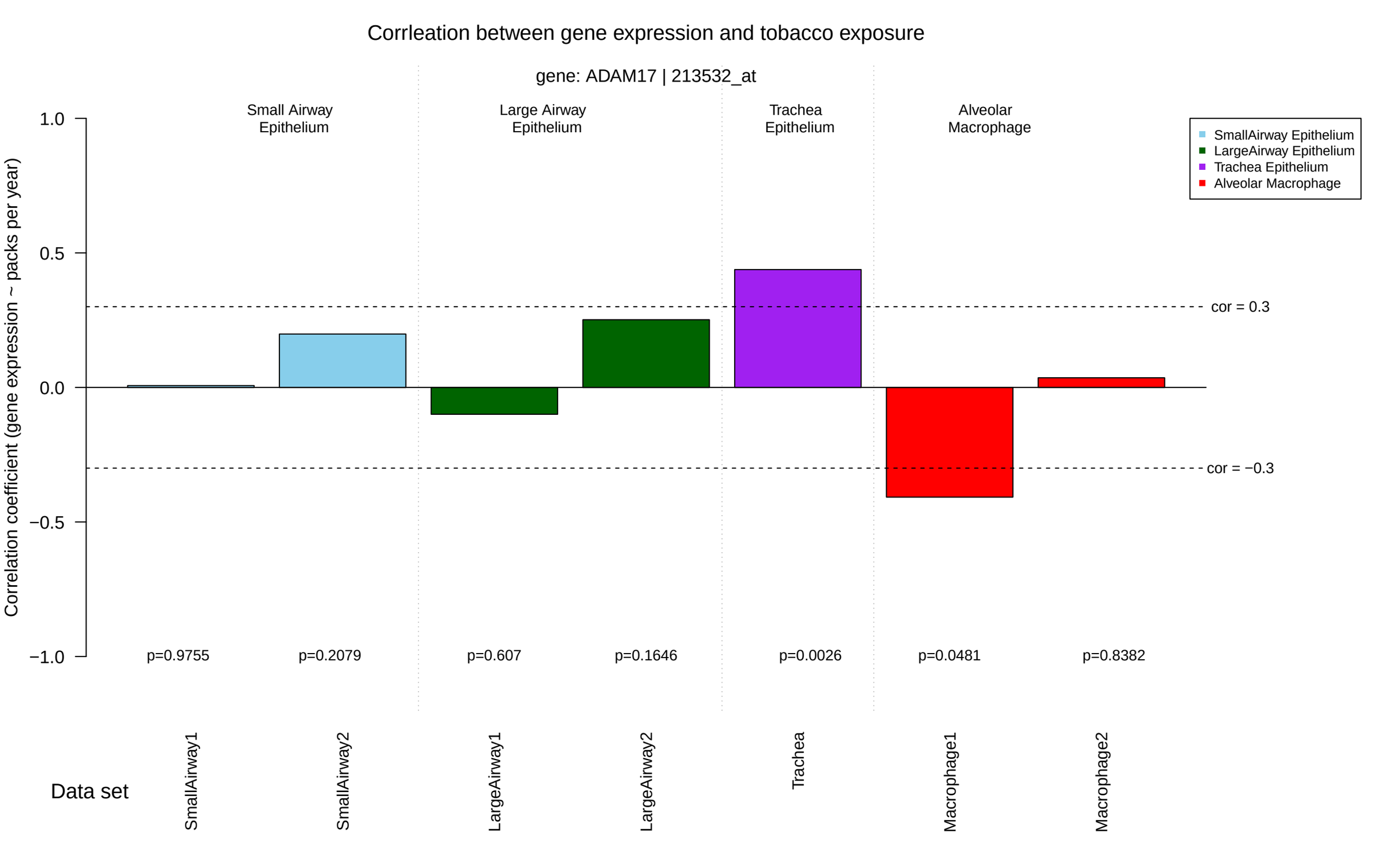

set: LargeAirway1 (p=0.9556)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0449)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0067)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2281)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2657)
ADAM17 | 213532_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAM17 | 213532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9755
p=0.2079
p=0.607
p=0.1646
p=0.0026
p=0.0481
p=0.8382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2